Prodigy Anesthesia Quick Review

Sympatholytics

What are the two major mechanisms by which alpha-2 receptor agonism reduces blood pressure?	Stimulation of peripheral presynaptic alpha-2 receptors inhibits catecholamine release, resulting in vasodilation. Stimulation of central alpha-2 receptors reduces sympathetic outflow and renin activity and produces sedation by decreasing circulating catecholamine levels. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 192.*
How are are clonidine, dexmedetomidine, and methyldopa classified pharmacologically?	They are all three alpha-2 receptor agonists *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 287-288.*
Should clonidine be continued through the perioperative period? Why or why not?	Clonidine should be continued throughout the perioperative period because abrupt discontinuation can result in rebound hypertension and tachycardia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 192.*
Does clonidine reduce blood pressure primarily by its action on peripheral alpha-2 receptors or its action on central alpha-2 receptors?	Agonism of the central alpha-2 receptors with subsequent reduction in sympathetic outflow is the primary mechanism by which it reduces blood pressure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 192.*
What is the primary use of dexemedetomidine?	The primary indication for dexmedetomidine is short-term sedation in critically ill patients. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 192.*
What are the potential side effects of dexmedetomidine?	Hypotension, bradycardia, oversedation, and delayed recovery. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 193.*
Does dexmedetomidine cause respiratory depression?	Dexmedetomidine does not produce significant respiratory depression. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 192.*
What are the typical intended drug effects of dexmedetomidine?	Dexmedetomidine provides anxiolysis, sedation, analgesia, and sympatholysis. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 192.*

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